Convergence, Tracking, and the LMS Algorithm

Step Size

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Key Concepts

1) The LMS algorithm updates the current iterate of the MMSE filter weights (or least-squares problem solution) by taking a step in the negative of the instantaneous gradient.

2) The size of the step taken impacts several aspects of LMS algorithm performance.

    a) The step size determines whether the LMS iteration is stable.

    b) The step size determines how quickly the iteration converges toward an optimal solution.

c) The step size determines the level of steady-state noise in the solution due to use of an instantaneous gradient.

d) The step size determines the ability of the LMS algorithm to track a time-varying optimal solution.

3) The MMSE filter (least squares problem) cost function is quadratic in the weights and thus is bowl shaped when $N = 2$.

4) The instantaneous gradient does not generally point to the optimum solution, i.e., the bottom of the bowl.

5) A large step size can lead to an unstable iteration as it causes the weights to climb up the sides of the bowl to larger and larger values of the cost function. A reasonable bound on μ for stability is

$$\mu < \frac{1}{\max_n \mathbf{x}^T[n]\mathbf{x}[n]}$$

Other bounds may be derived with various assumptions on the problem.

6) For very small step sizes:

    a) Decreasing μ results in an increased number of iterations for the solution to converge near the optimum.

b) Decreasing μ degrades the ability of the LMS algorithm to track a time-varying optimum solution.

7) Misadjustment refers to the effect of using an instantaneous gradient once the weights are near the optimum.

a) The noisy nature of the instantaneous gradient leads to weights that bounce around near the optimum rather than converge to and stop at the optimum.

b) The bouncing of the weights leads to a value of the cost function that is on average higher than the optimum value.

c) The degree of misadjustment decreases as the step size decreases.

8) Once stability is met, choosing μ involves trading off misadjustment (best when μ is small) for convergence and tracking ability (best when μ is larger).

9) The shape of the cost function determines the nature of convergence towards the optimum solution. For $N = 2$ the set of all weights with the same cost function value describes an ellipse.

a) The gradient of the cost function is normal to the contours of constant cost function value.

b) The major axis length is inversely proportional to the square root of the smallest eigenvalue of $\mathbf{R}$, while the minor axis is inversely proportional to the square root of the largest eigenvalue.

c) If the ratio of largest to smallest eigenvalue is large, the elliptical contours having constant cost function are very eccentric. The contours are closest together in the direction of the largest eigenvalue and thus convergence toward the optimum occurs fastest along this direction. Convergence can be slow along the direction of the smallest eigenvalue.

d) if the largest and smallest eigenvalues are equal, the contours of constant cost function values are circular and convergence occurs equally fast in all directions.

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MMSE Filtering or Least-Squares Problem

$$\min_{\underline{w}} \sum_{n=l_1}^{l_1+L-1} e^2[n] \quad ; \quad e[n] = d[n] - \underline{x}^T[n]\underline{w}$$

LMS (Gradient Descent) Algorithm

iterate on $\underline{w}$: $w[n]$

$$e[n] = d[n] - \underline{x}^T[n]\underline{w}[n]$$

$$\underline{w}[n+1] = \underline{w}[n] + \mu e[n]\underline{x}[n]$$

Step Size : μ

1) Stability
2) Rate of convergence
3) steady-state noise
4) Tracking a time varying solution

Consider a closely-related problem for insight

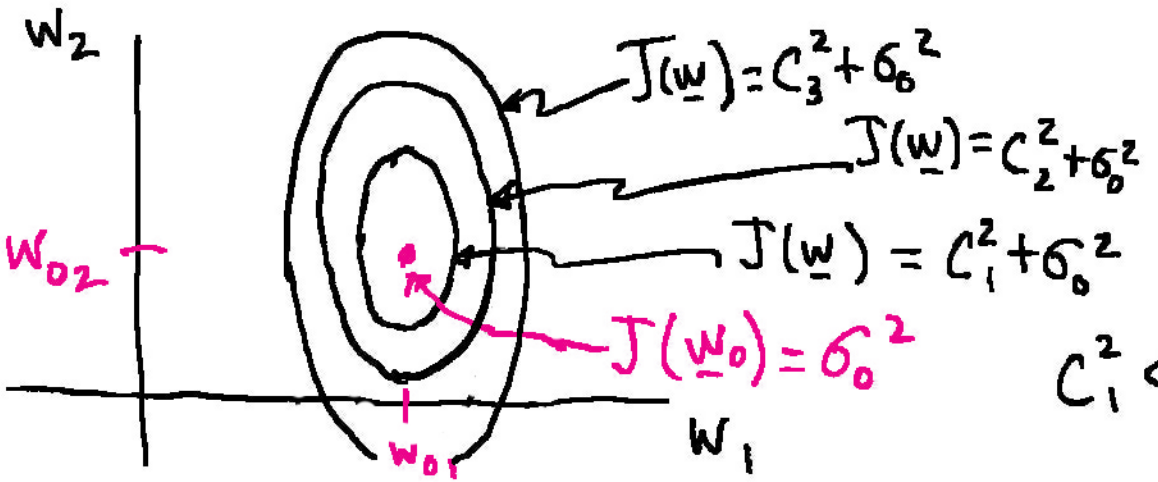
Cost function: $J(\underline{w}) = (\underline{w}-\underline{w}_o)^T \underline{R} (\underline{w}-\underline{w}_o) + \sigma_o^2$

$$\underline{w}_o = \underline{R}^{-1}\underline{p}$$

Special case: $N=2$, $\underline{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$, $\underline{R} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$, $\lambda_1 > \lambda_2$

$$J\left(\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}\right) = (w_1 - w_{o1})^2 \lambda_1 + (w_2 - w_{o2})^2 \lambda_2 + \sigma_o^2$$

$(w_1-w_{o1})^2 \lambda_1 + (w_2-w_{o2})^2 \lambda_2 = c^2$ is an ellipse



major axis (w_2): $2c/\sqrt{\lambda_2}$

minor axis (w_1): $2c/\sqrt{\lambda_1}$

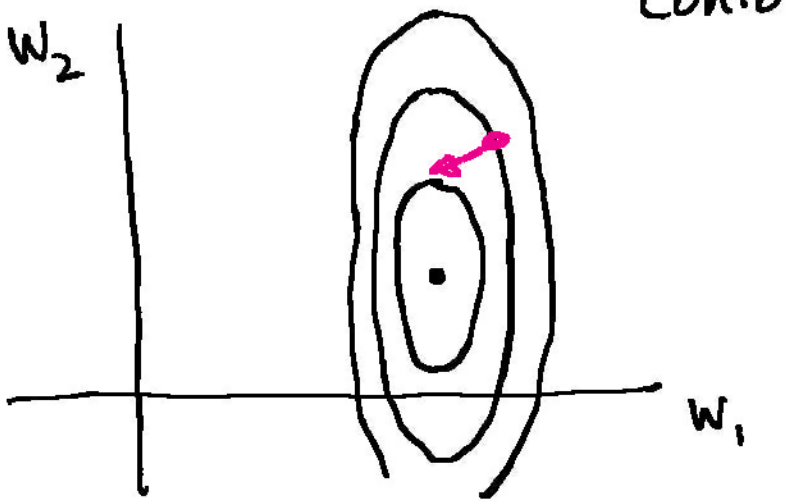
$c_1^2 < c_2^2 < c_3^2$

$J(\underline{w})$ "Bowl" shaped

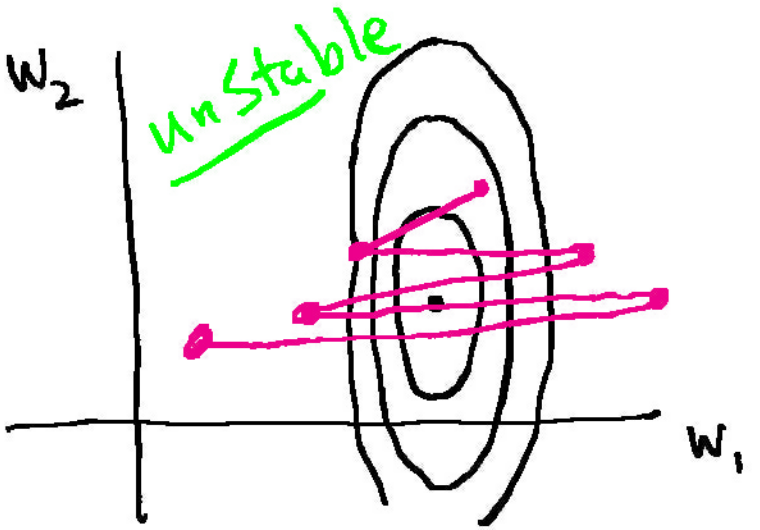
elliptical contours

major/minor: $\sqrt{\lambda_1/\lambda_2}$

Instantaneous gradient: "noisy" version of $\frac{\partial}{\partial \underline{w}} J(\underline{w})$



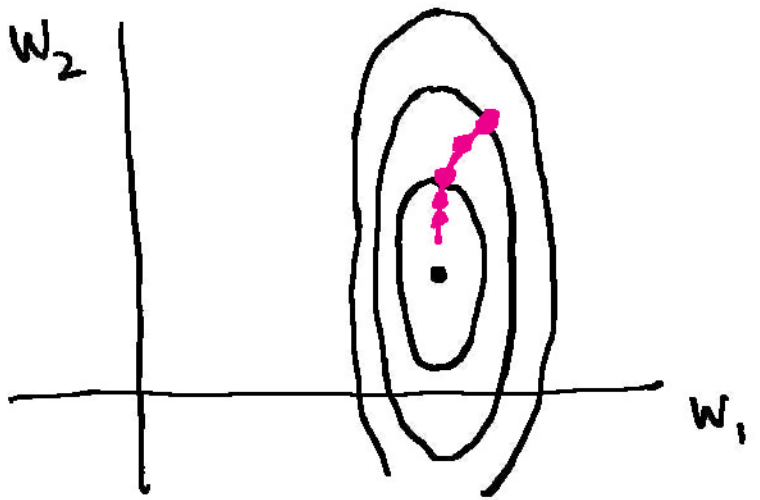
gradient of $J(\underline{w})$ at any point is normal to contours



large μ

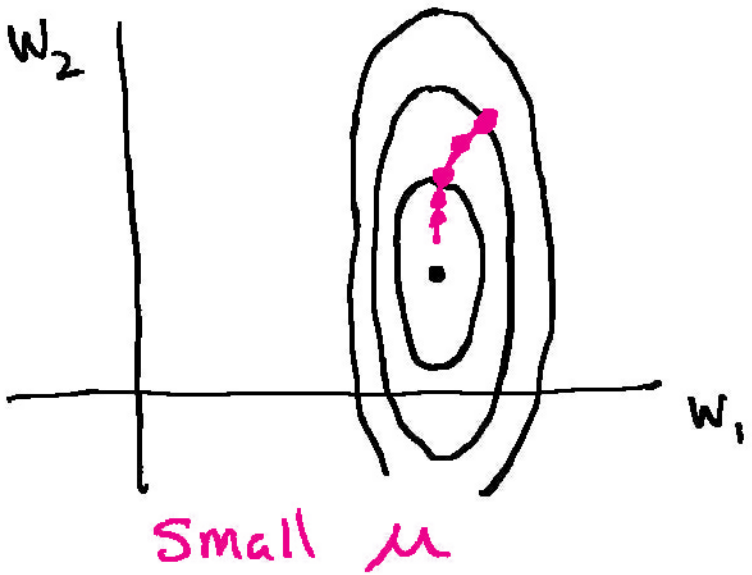


medium μ



small μ

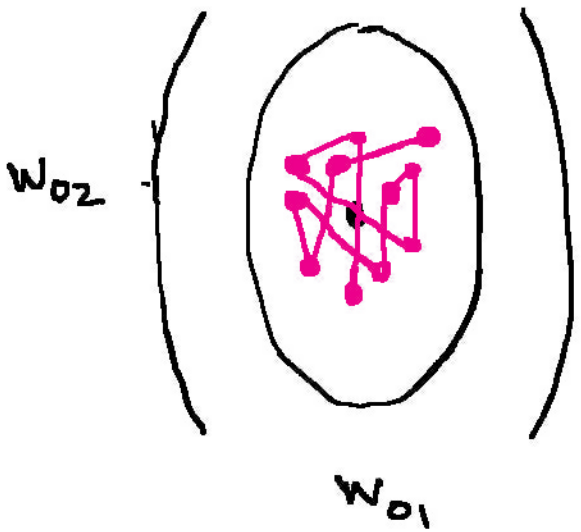
For stability: $\mu < \frac{1}{\max_n \underline{x}^T[n]\underline{x}[n]}$ (reasonable bound)



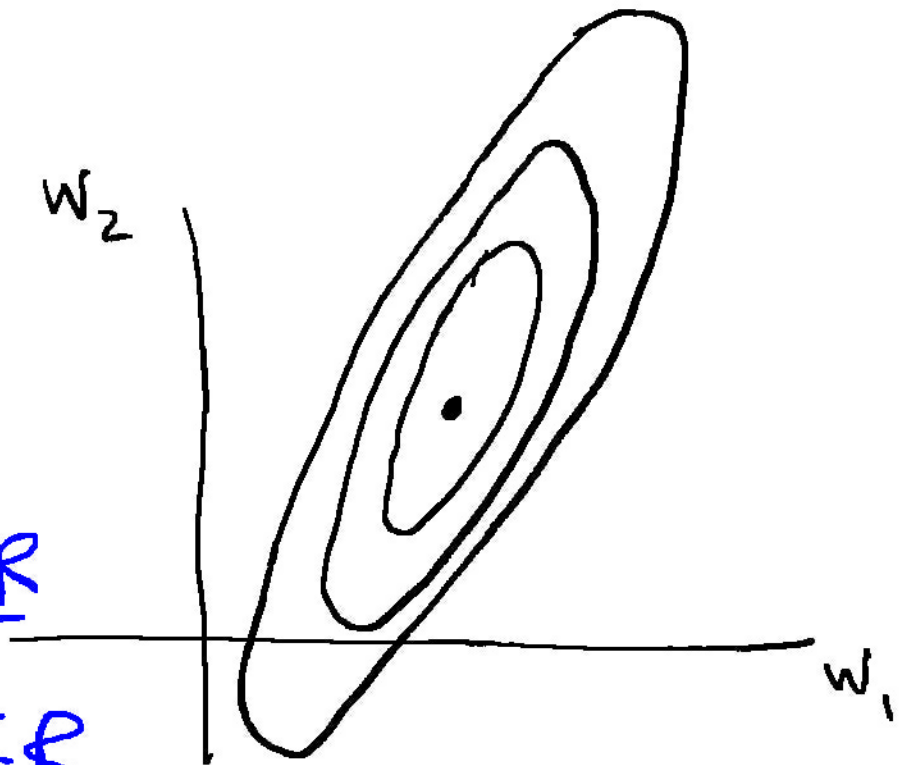
Observations –

1) tends to converge fastest along dimension with greatest power
2) Convergence can be slow for $\lambda_1 >> \lambda_2$ (very eccentric contours)
3) In general (small enough μ) convergence takes more iterations as μ decreases
4) Tracking a changing $\underline{w}_o[n]$ degrades as μ decreases

Misadjustment – noisy instantaneous gradient causes weights to bounce around near $\underline{w}_o$



misadjustment decreases as μ decreases



General non diagonal $\underline{R}$

orientation of major/minor axes: eigenvectors of $\underline{R}$

length of major/minor axes: eigenvalues of $\underline{R}$

Summary

1) Stability $\mu < \frac{1}{\max_n \underline{x}^T[n]\underline{x}[n]}$

2)

	μ small	μ large
convergence		✓
tracking		✓
misadjustment	✓	

3) Convergence in weights can be slow when $\frac{\lambda_{max}}{\lambda_{min}} >> 1$